Standard-PCA is performed on a centered dataset in $\mathbb{R}^3$. Two principal components are given below:

$$a = \frac{1}{2} \cdot \begin{bmatrix} 1 \\ \sqrt{2} \\ -1 \end{bmatrix}, \quad \overset{b}{\frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}}$$

Which of the following could be the third?

Let the 3rd PC be $\begin{bmatrix} a \\ b \\ c \end{bmatrix} = v$

$\boxed{v^T a = 0}$

$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/\sqrt{2} \\ 1/2 \end{bmatrix} = 0 \quad \text{——— (1)}$$

$\boxed{v^T b = 0}$

$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} = 0 \quad \text{——— (2)}$$

$$\left\| \begin{bmatrix} a \\ b \\ c \end{bmatrix} \right\| = 1 \quad \text{——— (3)}$$

$$\frac{a}{2} + \frac{b}{\sqrt{2}} - \frac{c}{2} = 0$$

$$\boxed{a + \sqrt{2}b - c = 0}$$

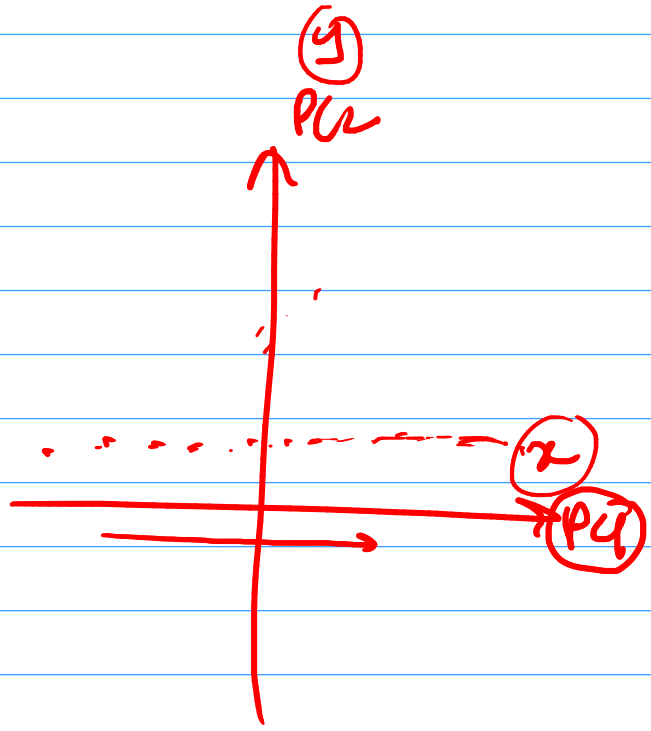
$$\frac{a}{\sqrt{2}} + \frac{c}{\sqrt{2}} = 0$$

$$\boxed{a = -c}$$

$$-c + \sqrt{2}b - c = 0$$

$$\sqrt{2}b = 2c$$

$$\boxed{b = \sqrt{2}c}$$



$$\sqrt{a^2+b^2+c^2} = 1$$

$$a^2+b^2+c^2 = 1$$

$$c^2+2c^2+c^2 = 1$$

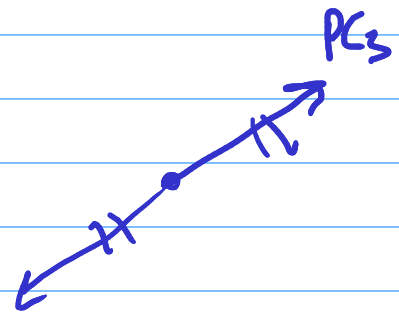
$$4c^2 = 1$$

$$c^2 = 1/4$$

$$\boxed{c = \pm 1/2}$$

$$a = -c \qquad b = \sqrt{2}c \qquad c = \pm\frac{1}{2}$$

$$c = \frac{1}{2} \quad a = \frac{1}{2} \quad b = \frac{1}{\sqrt{2}}$$



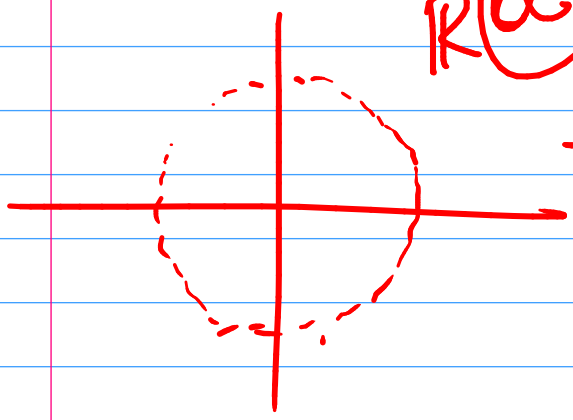
$$c = \frac{1}{2} \quad a = \frac{1}{2} \quad b = \frac{1}{\sqrt{2}}$$

$$\frac{1}{2}\begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \end{bmatrix} \text{ or } \frac{1}{2}\begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix}$$

Consider that 1000 data points belonging to d-dimensional space have a non-linear relationship. We apply kernel PCA to reduce the dimension of the data points and take the first k principal components. Can the value of k be larger than d?

1000 datapoints

$\mathbb{R}^d$ $\quad$ $\mathbb{R}^a$ $\quad$ $\mathbb{R}^p$



ϕ

$p >> d$

$\mathbb{R}^2 \rightarrow \mathbb{R}^7$ $\quad$ $4 = k > 2$

Let X be the data matrix of shape $d \times n$ with $d > n$ for a centered dataset. The eigenvector corresponding to the largest eigenvalue λ of $X^T X$ is α_1. What will be the first principal component of the dataset?

$$X_{d \times n} \quad \begin{bmatrix} | & | & & \\ c_1 & c_2 & \cdots & \\ | & | & & \end{bmatrix}_{d \times n}$$

$\boxed{x_i \in \mathbb{R}^d} \quad \forall i$

$$\boxed{w_1 = X\alpha_k} \qquad \|w_1\| = 1$$

$$w_1^T w_1 = w_1^T X \alpha_k$$

$$\|w_1\|^2 = w_1^T X \alpha_k$$

$$1 = (X\alpha_k)^T X \alpha_k$$

$$1 = \alpha_k^T X^T X \alpha_k$$

$$\boxed{1 = \alpha_k^T \underbrace{X^T X}_{K} \alpha_k}$$

non zero values of XX^T & X^TX are same

$$C = \frac{1}{n} XX^T$$

$$XX^T = nC$$

If eigen vector of C are $w_1\ w_2\ w_3 \ldots$

If eigen values of C $\lambda_1\ \lambda_2 \ldots$

$XX^T = nC$ eigenvector w_1 w_2 w_3

$n\lambda_1$ $n\lambda_2$ $n\lambda_3$

$\| \; x_1$

X^TX

eigen values $n\lambda_1$ $n\lambda_2$ $n\lambda_3$

eigen vector α_1 α_2 α_3

$\|\alpha_1\| = 1$

$$(X^TX)\alpha_1 = \lambda\alpha_1$$

$$\alpha_1^T(X^TX)\alpha = \lambda\,\alpha_1^T\alpha_1$$

$$\alpha_1^T(X^TX)\alpha = \lambda$$

$$\boxed{\alpha_1^T(X^TX)\alpha = \lambda}$$

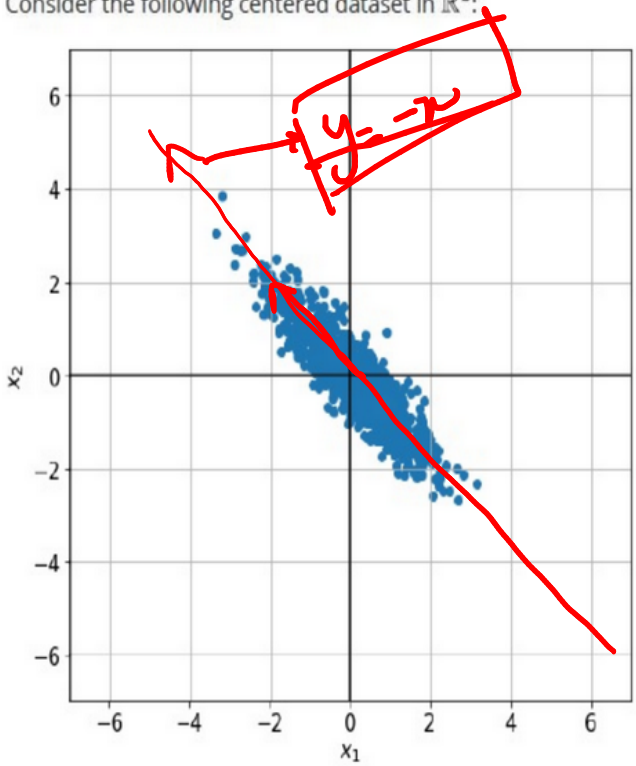
$$\alpha_1^T(X^TX)\alpha = 1$$

$$\frac{\beta_k}{\sqrt{n\lambda}} \to \lambda$$

$$\alpha_k = \frac{\alpha_1}{\sqrt{\lambda}}$$

$$W_1 = X\alpha_k = \frac{X\alpha_1}{\sqrt{\lambda}}$$

Consider the following centered dataset in $\mathbb{R}^2$:



Which of the following could be the first principal component? Recall that the first P.C is the most important one.

(a) $\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$ $\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$\frac{1}{n}\sum (x_i^T x_i)^2 = \boxed{\text{Error}} + \text{Variance}$$

+ve +ve

min max

$\boxed{\text{constant}}$

$5 \times 4 = x \times y$ $x \geq 0$ $y \geq 0$ $x, y \in I$

Consider a GMM with 3 components that is used to model a dataset that has 100 points. The EM algorithm is run on this dataset to estimate the parameters of the GMM. After convergence, some of the values of λ_k^i for the specific point x_{10} are given below:

$$\lambda_1^{10} = 0.2, \quad \lambda_2^{10} = 0.5$$

$k = 3$

$i \in \{1, \ldots 100\}$

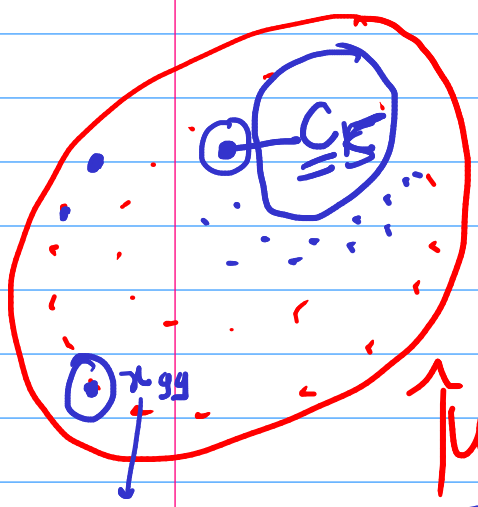
$x_{10} \rightarrow \boxed{\lambda_1^{10} \quad \lambda_2^{10} \quad \lambda_3^{10}}$

$\lambda_1^{10} + \lambda_2^{10} + \lambda_3^{10} = 1$

$\lambda_3^{10} = 1 - (0.2 + 0.5)$

$\boxed{\lambda_3^{10} = 0.3}$

D



$\lambda_5^{99} \simeq 0$

$\mu_k^{mm} = \dfrac{\sum_{i=1}^{100} \lambda_k^i x_i}{\sum_{i=1}^{100} \lambda_k^i}$

Consider a dataset with n points and a GMM with K components. If we fix λ and maximize for θ, what is our estimate for the mean of the kth component?

n points

k comp

$$\hat{\mu}_k^{MML} = \frac{\sum_{i=1}^{n} \lambda_k^i x_i}{\sum_{i=1}^{n} \lambda_k^i}$$

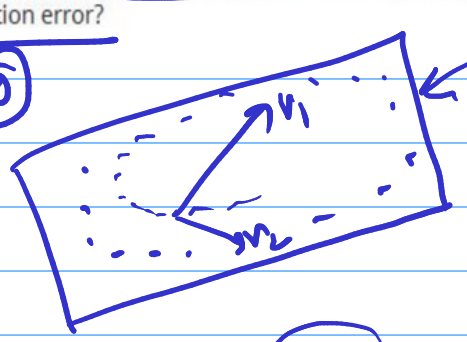
Consider a dataset of 20 points where the ith data-point is given by:

$$\mathbf{x}_i = a_i \cdot \underbrace{\begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix}}_{v_1} + b_i \cdot \underbrace{\begin{bmatrix} -1 \\ 1 \\ 3 \\ 0 \end{bmatrix}}_{v_2}$$

$$\boxed{v_1^T v_2 = 0}$$

where, a_i and b_i are real numbers such that $\sum_{i=1}^{20} a_i = \sum_{i=1}^{20} b_i = 0$. Standard PCA is performed on this dataset. If the top two principal components are retained and used to reconstruct the dataset, what is the reconstruction error?

0



20 SUBSPACE

$\mathbb{R}^4$

$$\sum_{i=1}^{20} x_i = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix} \sum_{i=1}^{20} a_i + \begin{bmatrix} -1 \\ 1 \\ 3 \\ 0 \end{bmatrix} \sum_{i=1}^{20} b_i$$

$$\mu_D = \frac{1}{n} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Consider the following prior for the parameter p of a Bernoulli distribution:

$$p \sim \text{Beta}(3, 2)$$

α β

The dataset observed is as follows:

$$\{1, 0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 1, 1\}$$

$n = 13$

What is $\hat{p}$, a point estimate for the parameter of the Bernoulli distribution, if we use the expectation of the posterior as the method of estimation? Enter your answer correct to three decimal places.

# of ones = 8

$$\boxed{\hat{p} = \frac{\alpha + n_{ones}}{\alpha + \beta + n}} = \frac{3+8}{5+13} = \frac{11}{18}$$

$= 0.61$

A dataset containing 200 examples in three-dimensional space has been transformed into a higher-dimensional space using a polynomial kernel of degree two. What will be the dimension of the transformed feature space?

p

$d = 3$

$p = 2$

$$\binom{d+p}{p} = {}^{d+p}C_p$$

$${}^5C_2 = \frac{5 \times 4}{2} = 10$$

Question Label Comprehension

Consider a centered dataset in $\mathbb{R}^{100}$ that has 1000 data-points. Call this dataset D_1. Standard PCA is performed on D_1 and the **scalar** projections of D_1 on the top 5 principal components are computed. Call the resulting dataset D_2.

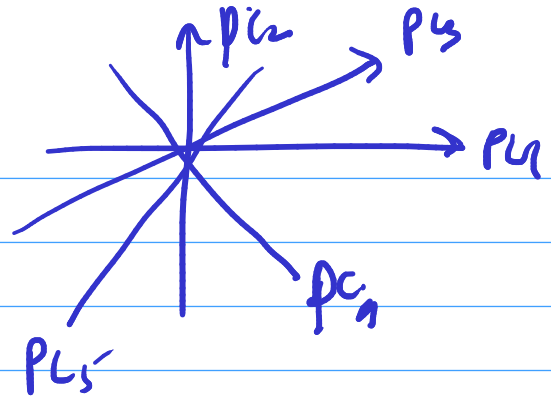
NOTE: D_2 is only made up of the scalar projections. All the principal components are thrown out after computing the scalar projections and are not a part of D_2.

D_1 $\mathbb{R}^{100}$

100 attributes $\in \mathbb{R}$

1000 datapoints

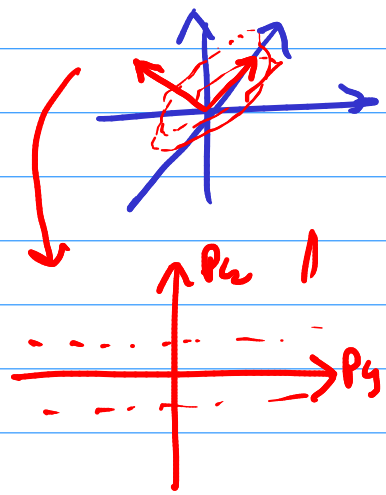
Each data-point in D_2 belongs to $\mathbb{R}^k$. What is the value of k?



$\mathbb{R}^5$

$\boxed{k=5}$

$\mathbb{R}^3$



If each real number occupies unit storage space in the memory, compute:

$$\frac{\text{size}(D_1)}{\text{size}(D_2)}$$

D_1 single datapoint has 100 attributes

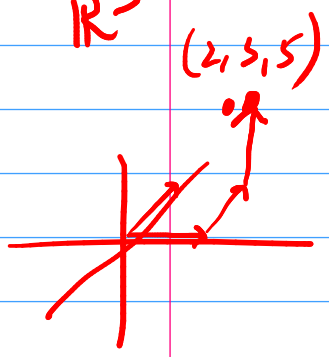
1000 " " 100×1000 attributes

10^5 attributes

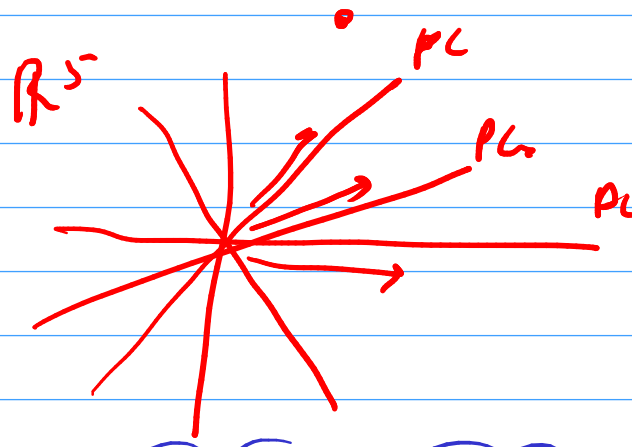
$\text{size}(D_1) = 10^5$ unit space

D_2 5 principal comp

$\mathbb{R}^3$



for each x_i



for each x_i → $\{x_i^T pc_1, x_i^T pc_2, x_i^T pc_3, x_i^T pc_4, x_i^T pc_5\}$

each new datapoint in $\mathbb{R}^5$ $\begin{bmatrix} - \\ - \\ - \\ - \\ - \end{bmatrix}$ will have 5 attributes

1000 datapoints with 5 attributes each

$$\text{size}(D_2) = 5 \times 1000$$
$$= 5000$$

Thou PC's $\in \mathbb{R}^{100}$

if not thrown out PC's $\quad \text{size}(D_2) = 5000 + 5$ PC's stored

$$= 5000 + 100 \times 5$$
$$= 5500$$

$$\boxed{\frac{\text{size}(D_3)}{\text{size}(D_2)} = \frac{105}{5000} = 20}$$

Assume that you have a dataset of five points $\{x_1, x_2, x_3, x_4, x_5\}$, all of which are non-negative. You hypothesise that the data points are iid random variables with the following density:

$$f(x;\lambda) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

What is the log-likelihood of this dataset under this distribution? ln represents the natural logarithm or $\log_e$.

$$\log(\mathcal{L}(D;\lambda)) = \log\left(\prod_{i=1}^{5} \lambda e^{-\lambda x_i}\right)$$

$$= \sum_{i=1}^{5} \log(\lambda e^{-\lambda x_i})$$

$$\log(\mathcal{L}) = \sum_{i=1}^{5} (\log\lambda - \lambda x_i \cancel{\log e})$$

$$\log(\mathcal{L}) = \sum_{i=1}^{5} (\log\lambda - \lambda x_i)$$

$$\frac{d\mathcal{L}}{d\lambda} = \frac{d}{d\lambda}\left(5\log\lambda - \lambda \sum_{i=1}^{5} x_i\right)$$

$$\frac{d\ell}{d\lambda} = \frac{5}{\lambda} - \sum_{i=1}^{5} x_i = 0$$

$$\frac{5}{\lambda} = \sum_{i=1}^{5} x_i$$

$$\boxed{\hat{\lambda} = \frac{5}{\sum_{i=1}^{5} x_i}}$$

$$\boxed{\hat{\lambda} = \frac{n}{\sum_{i=1}^{n} x_i}}$$

You are given the actual values of these observations:

$x_1 = 1, \quad x_2 = 2, \quad x_3 = 3, \quad x_4 = 4, \quad x_5 = 5$

What is the maximum likelihood estimate for λ? Enter your answer correct to three decimal places.

$$\hat{\lambda} = \frac{5}{15} = 0.33$$

A k-means++ algorithm with k = 3 was applied to following 2D points:

(0,0),(1,2),(3,1),(4,7),(-1,9),(4,-2)

(a,b) (c,d)

$|c-a| + |d-b|$

(0,0) is chosen as the first cluster mean.

Which point has the highest probability of being chosen as the 2nd cluster mean? Use the manhattan distance to compute the distances.

$\mu_1 = (0,0)$

	d
(1,2)	3
(3,1)	4
(4,7)	11
(-1,9)	10
(4,-2)	6

$\mu_2 = (4,7)$ 2nd Cluster mean

	d_{μ_1}	d_{μ_2}	$\min(d_{\mu_1}, d_{\mu_2})$
(1,2)	3	8	3
(3,1)	4	7	4
(1,9)	10	7	7
(4,-2)	6	9	6

$\boxed{\text{Maximum minimum distance}}$

Max among min colum = is (1,9)

Let $k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be a valid kernel. Is $\sin(x_1)\cos(x_2)k(x_1, x_2)$ a valid kernel? Here, $x_1, x_2 \in \mathbb{R}$?

for a valid kernel

$\boxed{\text{symmetric}}$ $\boxed{K(x_1, x_2) = k(x_2, x_1)}$

$\boxed{\text{Given:}}$ $K : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ is a valid kernel

Is $K' = \sin(x_1)\cos(x_2)\,K(x_1, x_2)$ a valid kernel

$x_1 = \pi/3 \qquad x_2 = \pi/6$

$$K'(\pi/3, \pi/6) = \sin(\pi/3)\cos(\pi/6)\,K(\pi/3, \pi/6)$$

$$= \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2}\,K(\pi/3, \pi/6)$$

$$= \frac{3}{4}\,K(\pi/3, \pi/6) \quad = A = \frac{3}{4}P$$

$$K'(\pi/6, \pi/3) = \sin(\pi/6)\cos(\pi/3)\,K(\pi/6, \pi/3)$$

$$= \frac{1}{2} \times \frac{1}{2}\,K(\pi/6, \pi/3) \quad = P$$

$\boxed{\text{NOT A VALID KERNEL}}$ $= \frac{1}{4}P$

Consider the following kernel:

$$k : R^2 \times R^2 \to R$$
$$k(x, y) = (x^T y)^2 + 1$$

Which of the following transformation mapping ϕ may correspond to the kernel k?

ϕ

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$k\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) = (x^T y)^2 + 1$$

$$= \left(\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right)^2 + 1$$

$$= (x_1 y_1 + x_2 y_2)^2 + 1$$

$$k(\underset{\sim}{x}, \underset{\sim}{y}) = \underbrace{x_1^2 y_1^2 + x_2^2 y_2^2 + 2 x_1 x_2 y_1 y_2 + 1}$$

$$k(\underset{\sim}{x}, \underset{\sim}{y}) = \phi^T(\underset{\sim}{x})\, \phi(y)$$

$$= \begin{bmatrix} x_1^2 & x_2^2 & \sqrt{2} x_1 x_2 & 1 \end{bmatrix} \begin{bmatrix} y_1^2 \\ y_2^2 \\ \sqrt{2} y_1 y_2 \\ 1 \end{bmatrix}$$

Consider a dataset consisting of 1000 samples in a 20-dimensional space. You apply Kernel PCA with a polynomial kernel of degree 3 to reduce the dimensionality of the data. Which of the following statements regarding Kernel PCA is true?

$\mathbb{R}^{20}$

trans dimension $\binom{20+3}{3} = \binom{23}{3}$

≈ 3500

if we choose 1000 PC's

then our new dataset will

belong to $\mathbb{R}^{1000}$

$\boxed{1000 > 20}$

Standard PCA has been performed on a centered dataset in $\mathbb{R}^3$. The first two principal components are given below:

$$\mathbf{w}_1 = \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{w}_2 = \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

x_i

Consider the data point in the dataset: $[2 \quad 1 \quad -1]^T$. (a, b) is the representation of this point in the coordinate system formed by the two principal components given above. The first and second coordinates correspond to PC-1 and PC-2 respectively.

new dataset lies in $\mathbb{R}^2$

(a, b) is rep in this new system

$x_i^T w_1 \qquad x_i^T w_2$

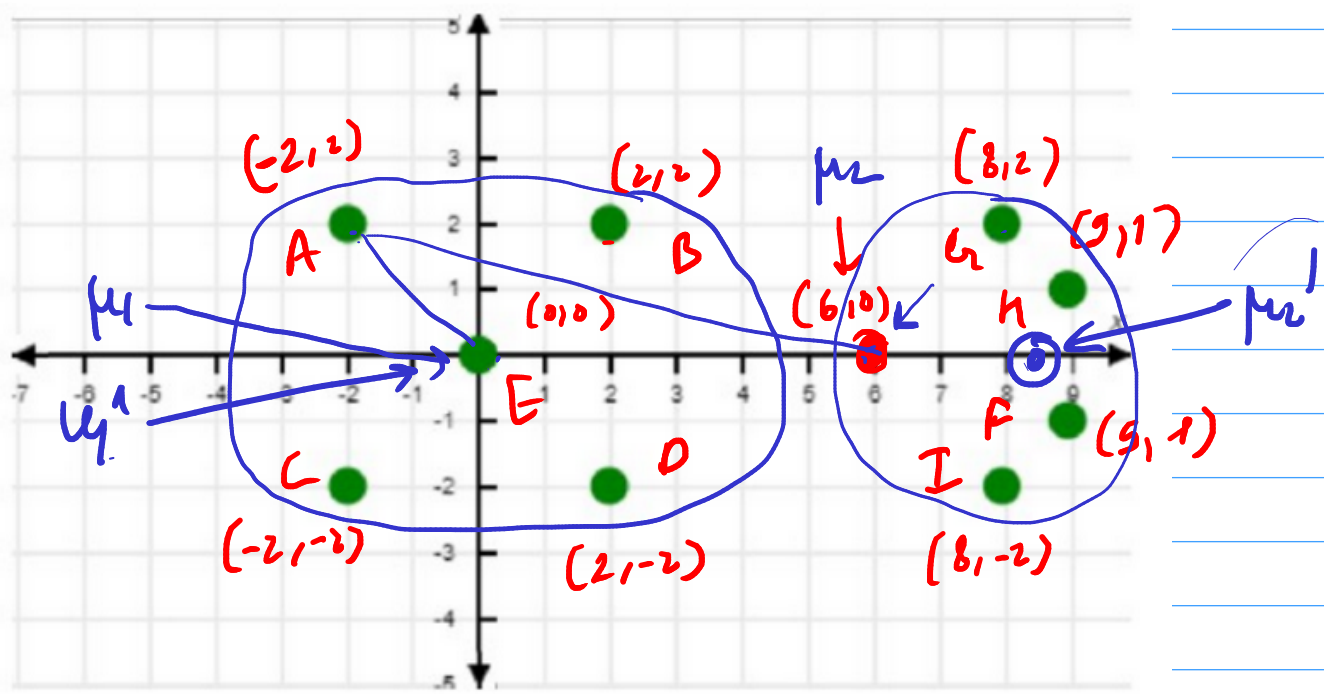
$$a = x_i^T w_1 = \begin{bmatrix} 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix} = \frac{2}{\sqrt{3}} + \cancel{\frac{1}{\sqrt{3}}} - \cancel{\frac{1}{\sqrt{3}}}$$

$$= \frac{2}{\sqrt{3}}$$

$$b = x_i^T w_2 = \begin{bmatrix} 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix} = \frac{2}{\sqrt{2}} - \frac{1}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}}$$

Consider the data points shown in the following image:



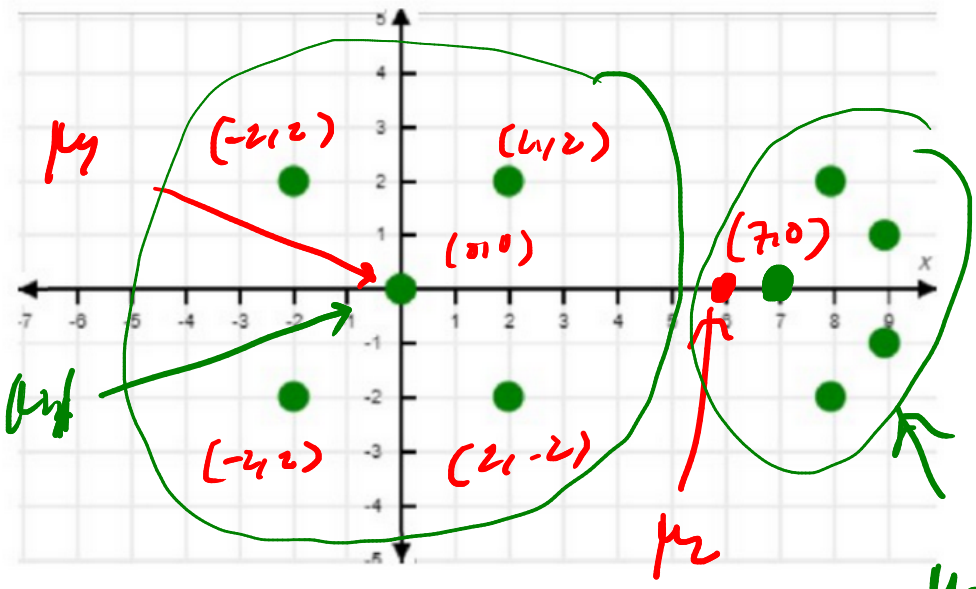
Based on the above data, answer the given subquestions.

Perform K-means clustering with $K = 2$ and initial cluster centers at (0, 0) and (6, 0). What are the final means of clusters after convergence?

$\mu_1 = (0,0)$ $\mu_2 = (6,0)$

$\mu_1' = (0,0)$ $\mu_2' = \left(\frac{34}{4}, \frac{0}{4}\right) = (8.5, 0)$

After introducing a new data point (7, 0) , the cluster centers were updated. Enter the sum of the updated x-coordinates for both cluster centers accurately, rounding your answer to two decimal places.



$$\mu_2' = \frac{1}{5}(34+7, 0)$$

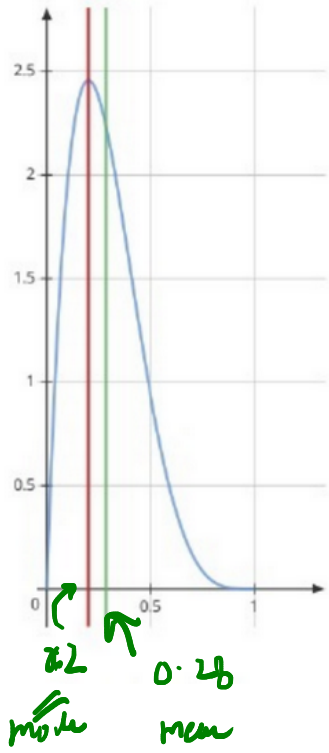
$$= \left(\frac{41}{5}, 0\right)$$

$$= (8.2, 0)$$

$$\boxed{0 + 8.2}$$

For the given Beta distribution, choose the correct option.

Beta(2, 5)



$$\text{mean} = \frac{\alpha}{\alpha+\beta} = \frac{2}{7}$$

$$\text{mode} = \frac{\alpha-1}{\alpha+\beta-2} = \frac{1}{5}$$

Consider a dataset with 100 total data points. Each data point is classified as either type A or type B. We model this using a Bernoulli distribution, where p is the probability of a data point being type A. If the maximum likelihood estimate (MLE) of p based on the dataset is 0.4, how many data points of type B are there in this dataset?

$$\hat{p} = \frac{\text{no. of a's}}{\text{no. of datapoints}} = 0.4$$

100

no. of a's $= 0.4 \times 100$

$= 40$

$$\boxed{\text{no. of b's} = 60}$$